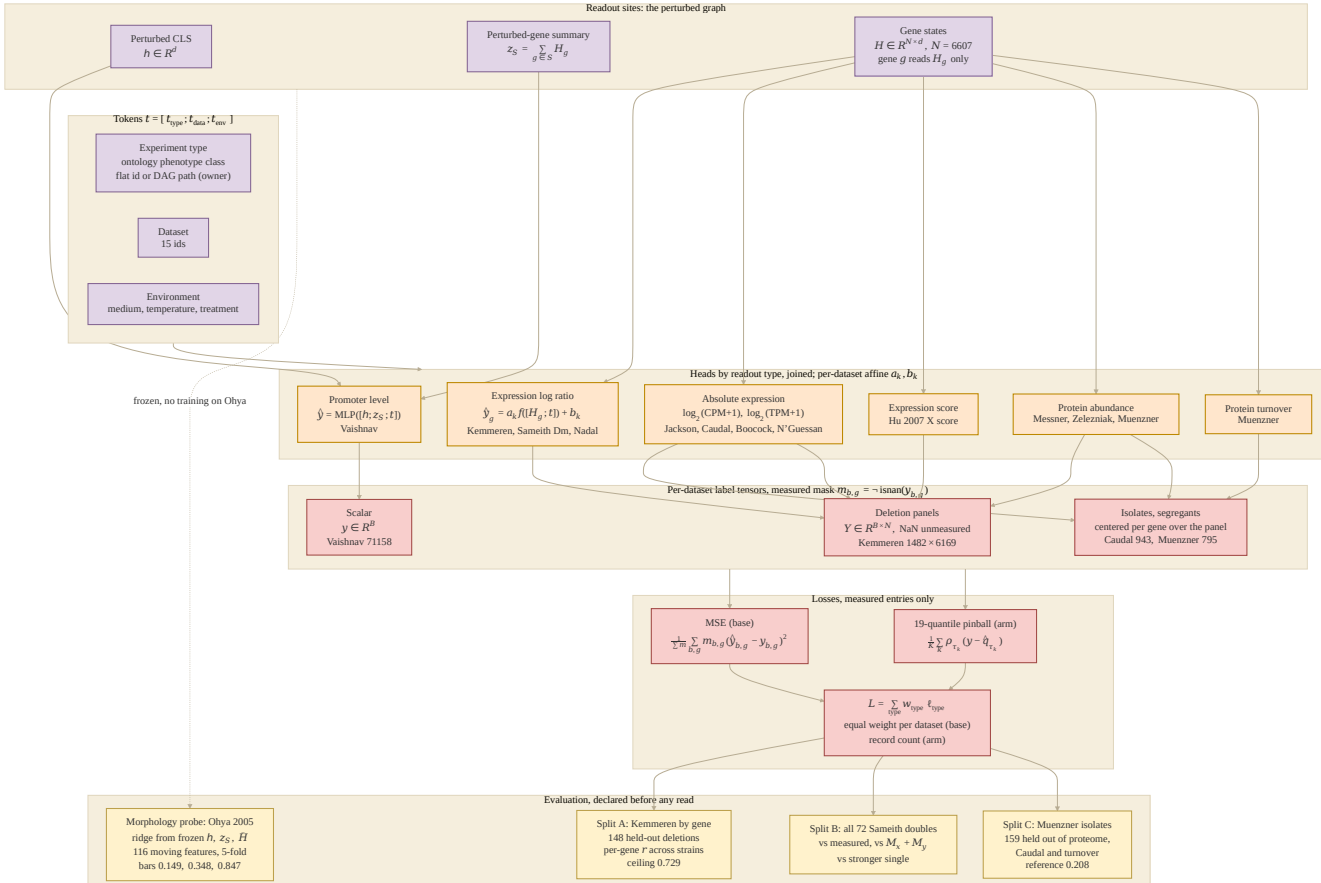

Readout sites: the perturbed graph
Perturbed CLS
$h \in R^d$
Perturbed-gene summary
$z_S = \sum_{g \in S} H_g$
Gene states
$H \in R^{N \times d}$, N = 6607
gene g reads H_g only
Tokens $t = [t_{type}; t_{data}; t_{env}]$
Experiment type
ontology phenotype class
flat id or DAG path (owner)
Dataset
15 ids
Environment
medium, temperature, treatment
frozen, no training on Ohya
Heads by readout type, joined; per-dataset affine a_k, b_k
Promoter level
$\hat{y}$ = MLP([h; z_S; t])
Vaishnav
Expression log ratio
$\hat{y}_g = a_k f([H_g; t]) + b_k$
Kemmeren, Sameith Dm, Nadal
Absolute expression
$\log_2$(CPM+1), $\log_2$(TPM+1)
Jackson, Caudal, Boocock, N'Guessan
Expression score
Hu 2007 X score
Protein abundance
Messner, Zelezniak, Muenzner
Protein turnover
Muenzner
Per-dataset label tensors, measured mask $m_{b,g} = \neg$ isnan($y_{b,g}$)
Scalar
$y \in R^B$
Vaishnav 71158
Deletion panels
$Y \in R^{B \times N}$, NaN unmeasured
Kemmeren 1482 × 6169
Isolates, segregants
centered per gene over the panel
Caudal 943, Muenzner 795
Losses, measured entries only
MSE (base)
$\frac{1}{\sum m} \sum_{b,g} m_{b,g} (\hat{y}_{b,g} - y_{b,g})^2$
19-quantile pinball (arm)
$\frac{1}{K} \sum_{\tau} \rho_{\tau_k} (y - \hat{q}_{\tau_k})$
$L = \sum_{type} w_{type} \ell_{type}$
equal weight per dataset (base)
record count (arm)
Evaluation, declared before any read
Morphology probe: Ohya 2005
ridge from frozen h, z_S, H
116 moving features, 5-fold
bars 0.149, 0.348, 0.847
Split A: Kemmeren by gene
148 held-out deletions
per-gene r across strains
ceiling 0.729
Split B: all 72 Sameith doubles
vs measured, vs $M_x + M_y$
vs stronger single
Split C: Muenzner isolates
159 held out of proteome,
Caudal and turnover
reference 0.208